Final Exam Project

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Abstract

This paper aims to identify the frequency and severity of traffic accident patterns in Los Angeles, utilizing a large historical dataset of traffic accident records. Traffic safety policies need to account for both accident frequency and factors influencing severity, like time, weather, and road infrastructure. This study analyzes the time of day, weather, and infrastructure at which traffic accidents occur most frequently and most severely. The study is conducted using descriptive analytics, spatial and temporal pattern analysis, and infrastructure-related risk analysis. The study utilizes the US Accidents dataset from February 2016 to March 2023 (Moosavi, 2023), filtering and preprocessing the data using Python and PostgreSQL, and analyzing the data using a star schema and interactive visualization using Tableau dashboards. The study found that traffic accidents occur most frequently during early morning hours and clear weather, whereas the severity of traffic accidents is higher during low-visibility conditions and complex infrastructure. Traffic safety interventions should target both high-frequency periods and high-severity contexts to effectively reduce overall accident risk and its impact.

Keywords: Traffic accidents, Severity, Los Angeles, Data visualization, Interactive dashboards, PostgreSQL, Star schema, Python, Tableau

1 Introduction

There is no doubt that transportation is vital to global economic and social development but at the same time, if not properly managed, it can have a high human and economic cost. Traffic accidents is a major public safety concern in Los Angeles. It is highly important to understand accident frequency and severity patterns in order to mitigate the problem. Based on a dataset of reported accidents from February 2016 to March 2023, this study examines temporal, environmental, and road infrastructure factors to identify conditions that increase risk.

The motivation is to provide data-driven insights to help city planners and traffic authorities prioritize interventions and allocate resources efficiently. Despite existing safety measures, high-severity accidents continue to occur, creating the need to assess when, where, and under what conditions accidents are most likely to happen.

In light of these considerations, the following research questions have been derived:

1. *Which months have the highest number of accidents and above-average severity?*
2. *Which hours of the day see the most accidents and whether severity peaks at these times?*
3. *Which weather and light conditions are associated with higher severity?*
4. *Which road features correlate with frequent and severe accidents?*

This study is limited to reported accidents in Los Angeles and relies on temporal, spatial, environmental, and road-related attributes available in the dataset. Minor or unreported incidents are not included, and results may not generalize to other regions. The report first presents the dataset and preprocessing methods, followed by analytical approaches and dashboards, then interprets results with actionable insights, and concludes with discussion and recommendations for traffic safety improvement.

2 Conceptual Framework

The analysis follows a staged framework, starting with data acquisition. It is followed by a quick data inspection on python and data preparation on postgreSQL through pgAdmin 4 which is important for improving data consistency and reduce noise (if any). Subsequently, feature engineering was applied to enable time based pattern analysis. Once the dataset was appropriately processed, data modeling followed using a star schema. Finally, the processed CSV files were exported from PostgreSQL and imported into Tableau, where the structured data were translated into interpretable patterns and analytical insights.

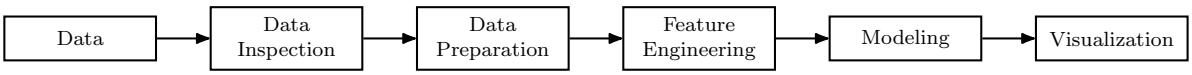

Figure 1: Conceptual Framework Diagram

These concepts will be further analyzed on the next section 3 Methodology

3 Methodology

3.1 Data Analysis Process

The analysis started by downloading the data from Kaggle. After a quick inspection, each record represents a single traffic accident with temporal, spatial, environmental, and road feature attributes. After the inspection, data preprocessing started with sql queries (Lecture 4 - Hussain, 2025b/ Lecture 5 - Hussain, 2025c). This process includes staging table creation for preserving the original dataset while enabling transformation, ensuring traceability and reproducibility. This is followed by duplicate checks, standardization of categorical variables, feature grouping, and systematic handling of missing values. Feature engineering was applied to extract temporal patterns, followed by construction of a star schema with fact and dimension tables. Finally, CSV exports were imported into Tableau for visualization and spatial-temporal analysis.

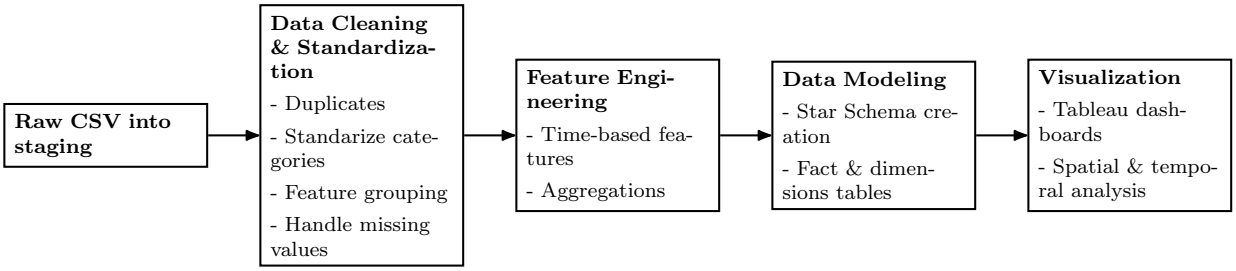

Figure 2: Data Analysis Process Diagram

3.2 Dataset Description

The dataset utilized on this paper is the *"US-Accidents (2016-2023)"* traffic accident dataset, publicly available on the Kaggle platform and originally compiled by Sobhan Moosavi. It is a countrywide dataset covering 49 states of the United States and contains records collected from February 2016 to March 2023 using multiple traffic event APIs, including data from transportation agencies, law enforcement, traffic cameras, and sensors within road networks. Each record in the dataset represents a single traffic accident and includes a variety of attributes related to time, location, weather conditions, road features, and accident severity. The dataset is provided as a single CSV file and contains millions of accident instances. In this study, the raw dataset comprised approximately 7.7 million records with 46 columns, while the processed dataset used for the analysis that includes only Los Angeles, has a shape of 156,491 rows x 27 columns (see Table 1).

3.3 Data Pre-Processing

The dataset was first filtered in Python because our research questions focus on Los Angeles, one of the 49 states included in the original dataset, where only the most relevant columns to the analysis were retained. A quick map inspection was taken place with seaborn and matplotlib library for an initial overview of spatial distribution. The dataset was then exported as a CSV file for database import.

In PostgreSQL, the CSV was imported, and a table was created matching the column names and structure of the exported file. The original dataset was retained in a staging table to enable safe data transformations while keeping the raw data information at the same time. Duplicate records were checked and none were found. Standardization was applied to text columns such as street to remove inconsistencies, and weather conditions were grouped into categories reflecting mild, moderate, or severe weather to simplify analysis (see Appendix - Table 1).

Then, missing values inspection followed and rows with combined missing weather information below 0.34% of the dataset were removed, while NULL entries in `street` and `weather_condition` were replaced with the placeholder "Not recorded." Temporal features were extracted from the accident timestamps, including day, month, year, day of the week, day of the year, week number, quarter, and local time, providing the basis for time-based pattern analysis. Accident times were aggregated to hourly intervals to simplify the data and make time-based patterns easier to analyze.

The following dimension tables were created from the staging table using distinct values (see Fig. 3: date_dim.csv, location_dim.csv, road_feature_dim.csv, weather_dim.csv). Each dimension table was assigned a surrogate primary key. These keys were then added to the staging table as foreign keys. The final fact table was created using the surrogate keys of the dimension tables together with numerical measures such as severity. The database was initially connected to Tableau for real time analysis and automatic updates in case they needed. However, due to performance limitations which are referring later on section 3.4.5, the live connection of the database with tableau, caused system crashes. So, in order to improve stability and ensure reproducible analysis, the data was exported into 5 different CSVs one for each dimension, which later used in Tableau for visualization and dashboard creation, allowing interactive exploration of spatial, temporal, and severity patterns.

3.4 Analytical Methods and Tools

Tableau was used for conducting data analytics and visualization, after the construction of the star schema in PostgreSQL (see Fig. 3 & Appendix Fig. 4). Fact table is linked to dimension tables, holding dimensions' primary keys as foreign keys (Lecture 03, Hussain, 2025a). This modeling approach improves query efficiency, and aligns the visualization

layer with the relational design as well.

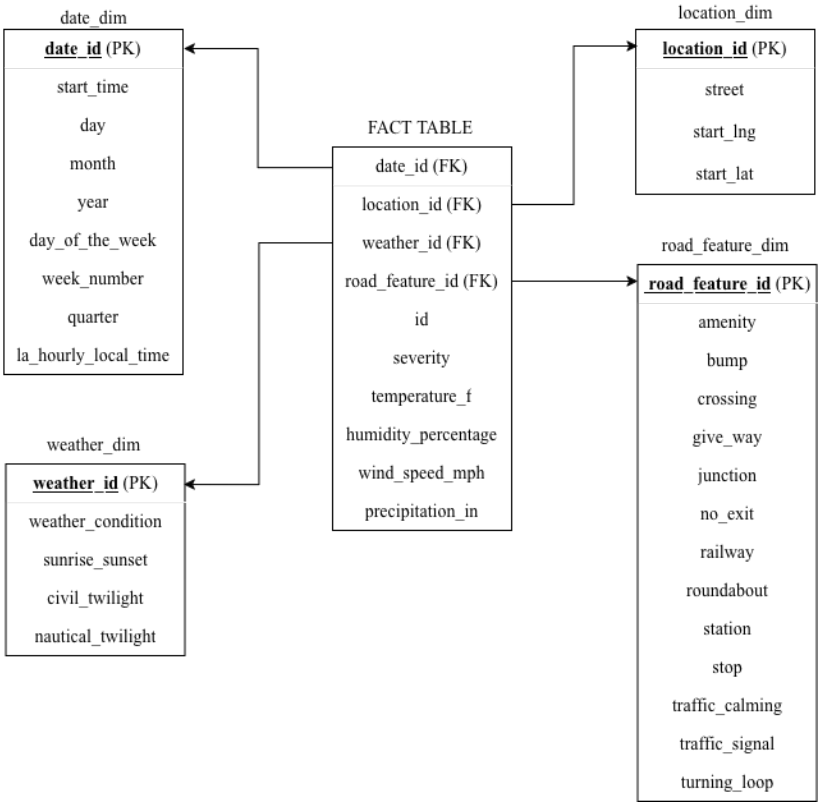

Figure 3: Star Schema

The analysis includes a descriptive and exploratory visual analytics approach. The goal was to identify spatial, temporal, environmental, and infrastructural patterns associated with traffic accidents and their severity. Dashboard's interactivity assists in exploration of identifying numerous of patterns.

3.4.1 Dashboard Structure and Design

The dashboards were designed using a top-to-bottom and left-to-right layout, prioritizing key indicators and high-level patterns before more detailed views. They also follow a minimal visual style with rectangle rounded-shaped edges. For the background, neutral gray tone has been used, while for containers (rounded rectangle boxes) white color has been kept for contrast (see Appendix Fig. 5). Navigation buttons have been integrated for switching between thematic dashboards focused on an overall overview, time, weather, and road infrastructure (see Appendix Fig. 6). This structure supports progressive analysis, from general patterns to specific dimensions. All dashboards are interactive through filters such as day, month, year, week no., hour, street name, and severity which are applied across views to ensure comparability of results (see Appendix Fig. 7 - 8). All these configurations have been taken in order to reduce cognitive load and keep reader's

attention on the data. Dashboards has been built in a way that through interactive filters, users can examine various more specific and in-depth questions, exploring patterns across time, location, weather, and road features according to their interests.

3.4.2 Visualizations and Metrics

There have been employed different kind of charts based on where the emphasis should be given (Lecture 10, p.4-8, Hussain, 2025d). `Geospatial map` to display accident distribution using latitude and longitude, with color encoding severity and size indicating accident frequency. Temporal visualizations including `line charts`, `heatmaps`, and `calendar` views to examine daily, monthly, yearly, weekly, and hourly accident patterns. `Bar plots` and `scatter plots` to analyze relationships between severity, weather conditions, and road features. `Heatmaps` to reveal interactions between weather conditions and lighting conditions.

Key performance indicators (KPIs) were included to provide contextual summaries: Total number of accidents, average accident severity, and percentage of high-severity accidents (severity level 3 and above). These KPIs provide a quick overview of the data and help interpret the patterns shown in the visualizations.

3.4.3 Calculated Fields

Continuing the analysis, the following calculated fields were created: An above-average severity indicator to identify periods or categories with higher severity than the average, which would help later on color segmentation (Lecture 10, p.11, Hussain, 2025d). A high-severity accident rate, focusing on severity levels with greater impact, important to define one of the three KPIs referred to above. These calculations allow severity to be analyzed both relative to isolation and to overall trends.

3.4.4 Color Encoding and Visual Emphasis

A custom four-level color palette was created and applied across all visualizations of the dashboard (see Appendix Fig. 9) to clearly distinguish the four levels of accident severity. The light gray color (#d3d3d3) represents the lowest severity level. A slightly darker gray (#b1a7a6) was used for the second severity level to indicate elevated risk. These gray tones were selected for lower severity levels to indicate situations without serious consequences and to avoid overemphasizing minor accidents when that's the case. Same logic for the tones applies to the red color (#e5383b and #ba181b). These tones were used for higher severity levels, as red is commonly associated with danger and critical events in traffic safety contexts.

In visualizations where severity is not the primary measure, the same color palette is used to represent the number of accidents, with lighter tones indicating lower frequencies

and darker tones indicating higher concentrations. The choice of keeping the same color palette maintains visual consistency across the dashboard with that color always to reflect increasing intensity. The specific color in each chart is clarified through titles, axis, and detailed information by hovering over individual marks depending on the chart.

3.4.5 Dataset Limitations

The processed dataset is large and makes real-time analysis in visualization tools computationally intensive. Directly connecting Tableau to the database was attempted, but the software could not efficiently process the rows with complex calculations and multiple interactive filters. This is the reason that CSV files used as the data source in Tableau. However, this limits the ability of performing real-time queries on the full dataset. It loses the advantage of being able to make continuous updates on the database, but for large-scale exploratory dashboards, CSV files is often more practical.

Moreover, the dataset includes only reported accidents. Unreported incidents are not captured and it misses data for certain days, which are due to network connectivity issues during author's data collection.

3.4.6 Data Analytics: Alternate Methods and Cost - Benefit Analysis

An alternative approach could have been to retrieve the accident data directly using traffic APIs. This would allow access to the most up-to-date or custom data and avoid dependency on ready datasets, but it would come with the catch of less available data due to time restriction. Instead, using the Kaggle dataset combined with SQL and Tableau offered a balance were the data were already compiled and cleaned to some extent, as well as manageable for large-scale aggregation. SQL provided accurate preprocessing and feature engineering, while Tableau provided interactive exploration and visualization.

4 Results

This section presents the results of the exploratory analysis of traffic accidents in Los Angeles. Findings are based on patterns identified around accident frequency, severity, weather conditions, daylight, and road infrastructure.

4.1 Meaningful Facts

The results that will be discussed on this subsection, are basically descriptive observations based on the dashboard for each one of the research questions. However, results vary when filtering.

Firstly, analysis of the dataset for the whole period of 2016-2023 shows that December had the highest number of traffic accidents, 15,538 in number, whereas July recorded the lowest number of accidents of 10,349. Regarding severity, all the months from March until August consistently exceed average severity levels, with July having the highest average severity of approximately 2.3 out of 4.

Secondly, it is obvious that the highest number of accidents occurs between 4:00 AM and 8:00 AM, corresponding to early morning traffic. In contrast, the number of accidents during these hours tend to have lower severity than the average. The most severe ones are observed during hours with lower overall accident counts.

Thirdly, cloudy conditions are associated with the highest average severity, followed by fog/mist, not recorded values, and clear conditions. Considering accident frequency, accidents occur most often during clear conditions in full daylight, followed by clear conditions at night, cloudy conditions in daylight, and cloudy conditions at night.

Fourthly, road segments with only an exit feature are associated with the highest number of accidents, indicating that these areas experience frequent but generally less severe accidents. On the other hand, road segments combining exit, railway, and station features are linked with the highest average severity.

4.2 Actionable Insights

The observed patterns in the dashboard provide a clear guidance for intervetions in order to reduce both the number of accidents and their severity.

During high-frequency but low-severity periods, such as early morning hours between 4:00 AM and 8:00 AM, it is recommended to maintain existing traffic safety measures to prevent accident numbers from increasing.

For periods or conditions with moderate to high severity, targeted interventions are required. For example, enhanced enforcement, signage, or visibility improvements.

Furthermore, measures regarding road infrastructure should be considered. Segments combining exit, railway, and station features that have the most severe accidents will be benefit from engineering or control measures to reduce the risk of serious accidents. Similarly, warning signs and speed management can help reduce these incidents on weather and light sensitive occasions.

Lastly, for an effective accident risk mitigation, resources should be carefully allocated. For instance, uniform policies across all times and locations should be avoided, as they are unlikely to address the most critical risks as they should be.

On section 5 Discussion, will be further analyzed the possible reasons behind the observed patterns and their implications for traffic safety.

4.3 Valuable Outcomes

The above insights can inform traffic safety planning across different time horizons. Short-term, they could improve decision-making based on periods and locations with high accident frequency and severity, providing driver awareness of high-risk conditions. Over the mid term, the findings can support infrastructure planning and policy adjustments. In the long term, results can inform predictive and preventive systems that combine historical patterns with real-time data. In this way more effective and data-driven traffic safety policies will guide urban planning initiatives in Los Angeles.

5 Discussion

As discussed above, December is the peak month of accidents. This peak is likely related to seasonal changes in traffic volume and driving behavior, like higher congestion, time pressure etc. At this time, social events during holidays take place, with more distractions for the drivers and occasional risk-taking, which can may also includes alcohol consumption. Environmental conditions appear to play a stronger role in accident severity than in accident frequency.

The time periods with the highest accident frequency are not strongly associated with higher severity. Although most accidents occur between 4:00 AM and 8:00 AM, the average severity during these hours is close, but below the overall mean. This demonstrates that high-frequency periods do not necessarily correspond to high risk in terms of injury severity. Distinguishing between frequent incidents and high severity conditions enables more targeted and proportionate road safety interventions in Los Angeles.

Regarding light conditions, most of the accidents occur under clear daylight, while higher severity is associated with reduced visibility, during cloudy and fog or mist conditions. So, frequency is influenced by driving behavior, insufficient road infrastructure or other conditions. However, severity is not driven by frequency alone, but is influenced by environmental conditions that limit visibility and reaction time.

Focusing on road infrastructures, most accidents occur on segments with only an exit nearby and no additional traffic features. In contrast, roads combining exits, railways, and stations are associated with higher severity, likely due to the increased complexity of these environments. This combination makes collisions more severe and government should take measures to reduce this risk, for example, by providing awareness to the public.

5.1 Implications, Limitations, and Future Work

The findings underscore the importance of combining temporal, environmental, and infrastructure factors under traffic safety context. Future studies could explore predictive

models both for frequency and severity using these dimensions and integrate real-time data.

The results also provide actionable guidance for traffic authorities. From seasonal safety campaigns, traffic control improvements, to dynamic measures that respond to weather and light conditions, such as automatic speed limit adjustment, install or turning on existing reflective markers when natural light is low, can be taken into consideration.

Turning to limitations, the dataset includes only reported accidents, neglecting minor or unreported incidents. Weather and road features had missing values, which could not be imputed and this may have affected results.

The study demonstrates the utility of integrating SQL preprocessing with Tableau dashboards for descriptive traffic safety analysis. It also highlights challenges in balancing dataset size, visualization performance, and interpretability.

Some of the next steps could include predictive modeling using machine or deep learning, update the original dataset with up-to-date data using traffic APIs, and expand analysis to include other cities or states for comparative insights.

6 Conclusion

This study has conducted and investigated patterns about accident frequency and severity, influenced by factors like time, weather, and road infrastructure on Los Angeles from 2016 to 2023 with high-severity incidents to happen under adverse conditions or complex road layouts. Using SQL preprocessing and Tableau dashboards, allowed interactive exploration of spatial and temporal patterns. Despite limitations, the paper demonstrates the value of combining structured data and visual analytics, providing actionable insights for targeted initiatives in order to reduce the collusion numbers.

References

Hussain, A. (2025a). Lecture 03 - operational and analytical rdbms.

Hussain, A. (2025b). Lecture 04 - structure query language (sql).

Hussain, A. (2025c). Lecture 05 - advanced sql end 2 end analytical db implementation.

Hussain, A. (2025d). Lecture 10 - visual design considerations hands on full dashboard design tableau power bi.

Moosavi, S. (2023). Us accidents (2016 - 2023). https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents

7 Links

Los_Angeles_Traffic_Accidents_Dashboard

Appendix

Column	**type**
id	TEXT PRIMARY KEY
severity	INT
start_time	TIMESTAMP WITH TIME ZONE
start_lat	DOUBLE PRECISION
start_lng	DOUBLE PRECISION
street	TEXT
temperature_f	FLOAT
humidity_percentage	FLOAT
wind_speed_mph	FLOAT
precipitation_in	FLOAT
weather_condition	TEXT
amenity	BOOLEAN
bump	BOOLEAN
crossing	BOOLEAN
give_way	BOOLEAN
junction	BOOLEAN
no_exit	BOOLEAN
railway	BOOLEAN
roundabout	BOOLEAN
station	BOOLEAN
stop	BOOLEAN
traffic_calming	BOOLEAN
traffic_signal	BOOLEAN
turning_loop	BOOLEAN
sunrise_sunset	TEXT
civil_twilight	TEXT
nautical_twilight	TEXT

Table 1: Data types

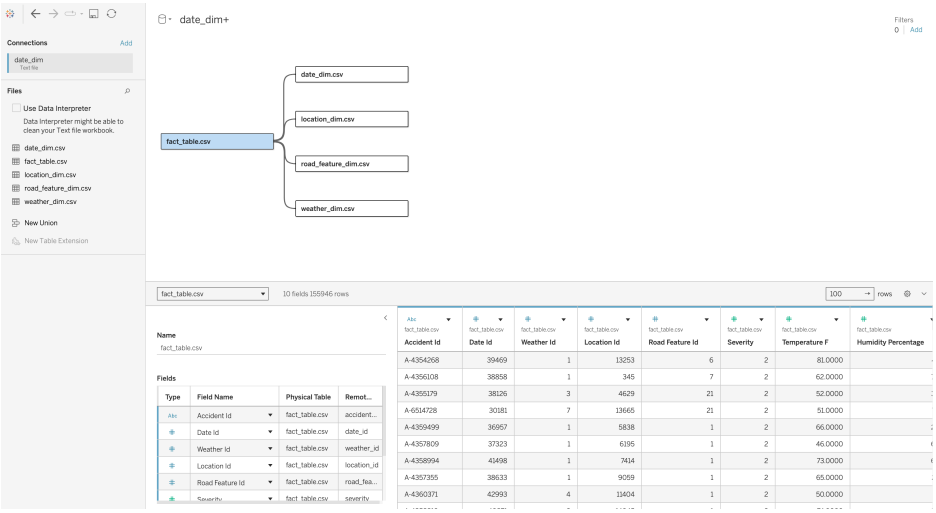

Figure 4: Tableau Data Source

Figure 5: Overview Layout

Figure 6: Navigation Buttons

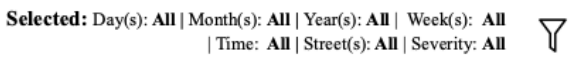

Figure 7: Filters 1

Figure 8: Filters 2

Figure 9: Custom Color Palette